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Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
  - three syntheses done with 0.02M sulfuric acid
    - first product not representative, not used further
    - second and third batch converted (see below)
  - four syntheses done with 0.10M sulfuric acid
    - very similar observations and outputs
    - products of these have been combined/homogenized
    - part of the combined lots has been converted

- Canadian Rockies Hemp arrived and cleared customs
  - several standard synthesis batches finished, seven of these (11.2 g) combined and homogenized
  - additional experiments done with higher concentrated sulfuric acid (1.5%, 2.0%, 5.0%)
  - one experiment performed under stirring, using cut up hemp fibers, product is finer and more homogeneous
  - one experiment performed with sulfuric acid pre-treatment of hemp (95 °C)

- Larger vessel reactivated, production ongoing (approx. 15 g output per batch)

- Typical moisture content 5% by KFT. Further drying at 100 °C under nitrogen stream decreases KFT result to 2%. Relevance unclear – being tested in step 2.

2. Biochar analysis

- Sample of combined biochar (milled, MB2946X_gem) analyzed by SEM and Raman for reference

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
  - biochar ground manually for first experiment (TB1131)
  - laboratory mill used for second experiment (TB1132)
  - products not yet characterized
  - no serious handling issues

- two experiments performed with much gentler heating ramp (3 °C/min, TB1133/1134)

- better handling but no other apparent difference
- one experiment performed at 5 °C/min (MB2946)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 1 h: minimal yield (MB2947A)
  - 800 °C, 2 h: minimal yield (MB2947B)
  - 800 °C, 3 h: no output (MB2947C)
  - ➔ Inertion apparently insufficient and material likely incinerated
  - 800 °C, 3 h repeat, much higher Ar stream: 38 mg output obtained (MB2947D)
  - 800 °C, 1 h, nitrogen inertion: 60 mg output (TB1135-1)
  - 800 °C, 1 h, increased nitrogen inertion: 70 mg output (TB1135-2)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 2 h, inertion equivalent to TB1135-2: 70 mg output (MB2952B)
  - 800 °C, 3 h, inertion equivalent to TB1135-2: 50 mg output (MB2952C)
    - ➔ smoldering observed upon contact of the cold material with air (see video)
  - 800 °C, 3 h, inertion equivalent to TB1135-2: 60 mg output (MB2952C2)
    - ➔ material wetted with water before/during removal from the apparatus

- Oxygen sensor shows complete inertion of apparatus within 3 minutes at selected nitrogen stream (equivalent to TB1135-2)

- 2 g of combined biochar milled with 3 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 1 h, standard inertion: Quartz tube shattered, only 7 mg output (MB2955A)
  - 800 °C, 1 h, standard inertion (repeat of MB2955A): 90 mg output (MB2955A2)
  - 800 °C, 2 h, standard inertion: 80 mg output (TB1136)

- 2 g of combined biochar milled with 4 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 1 h, standard inertion: 80 mg output (MB2962A)
  - 800 °C, 2 h, standard inertion: 90 mg output (MB2962B)

- 0.6 g of biochar (cut up, hydrolyzed under stirring) milled with 0.6 g of KOH and used for two batches on 0.2 – 0.3 g scale
  - 800 °C, 1 h, standard inertion: 70 mg output (MB2963A)
  - 800 °C, 2 h, standard inertion: 120 mg output (MB2963B)

- 2 g of combined biochar milled with 8 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 1 h, standard inertion: 60 mg output (TB1137-1)
  - 800 °C, 2 h, standard inertion: 60 mg output (TB1137-2)
  - 800 °C, 1 h, standard inertion: 70 mg output (TB1137-3)

- 2 g of biochar lot MB2959 milled with 12 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion: 24 mg output (MB2965A)
  - 800 °C, 2 h, standard inertion: 13 mg output (MB2965B)
  - 800 °C, 2 h, standard inertion (oven B): 7 mg output (MB2965B2)

- 2 g of biochar lot MB2959 milled with 8 g of KOH and used for several batches
    - 800 °C, 1 h, standard inertion (oven B): 52 mg output (MB2966A)
    - 800 °C, 2 h, standard inertion: 46 mg output (MB2966B)
- 1.2 g of biochar lot MB2964 milled with 4.8 g of KOH and used for several batches
    - 800 °C, 1 h, standard inertion (oven B): 57 mg output (MB2967A)
    - 800 °C, 2 h, standard inertion: 42 mg output (MB2967B)

- 1.4 g of biochar lot MB2957 (from 1.5% H_2SO_4) milled with 2.1 g of KOH and used for several batches
    - 800 °C, 1 h, standard inertion (oven B): 63 mg output (MB2970A)
    - 800 °C, 2 h, standard inertion: 19 mg output (MB2970B)

- 1.4 g of biochar lot MB2958 (from 2% H_2SO_4) milled with 2.1 g of KOH and used for several batches
    - 800 °C, 1 h, standard inertion (oven B): 54 mg output (MB2971A)
    - 800 °C, 2 h, standard inertion: 66 mg output (MB2971B)

- 1.4 g of biochar lot MB2961 (from 5% H_2SO_4) milled with 2.1 g of KOH and used for several batches
    - 800 °C, 1 h, standard inertion (oven B): 88 mg output (MB2972A)
    - 800 °C, 2 h, standard inertion: 57 mg output (MB2972B)

- 1.4 g of biochar lot MB2959 milled with 2.1 g of KOH and used for two identical batches (reproducibility experiment)
    - 800 °C, 1 h, standard inertion (oven B): 42 mg output (MB2973A)
    - 800 °C, 1 h, standard inertion: 90 mg output (MB2973B)

- 1.5 g of biochar lot MB2959 milled with 2.3 g of KOH and used for two batches (milling time experiment)
    - milling time 5 min, 800 °C, 1 h, standard inertion (oven B): 78 mg output (MB2974A)
    - milling time 1 min, 800 °C, 1 h, standard inertion: 81 mg output (MB2974B)

- 1.5 g of biochar lot MB2959 milled with 2.3 g of KOH and used for two batches (milling time experiment)
    - milling time 5 min, 800 °C, 2 h, standard inertion (oven B): 59 mg output (MB2975A)
    - milling time 1 min, 800 °C, 2 h, standard inertion: 80 mg output (MB2975B)

- 2.0 g of biochar lot MB2959 milled with 3.0 g of KOH and used for two batches (pellet pressing experiment)
    - 0.6 g pellet, 800 °C, 1 h, standard inertion (oven B): 83 mg output (MB2976A)
    - 0.6 g pellet, 800 °C, 2 h, standard inertion: 108 mg output (MB2976B)

- Biochar/KOH mixture from MB2976 used for new reproducibility experiment (analogous to MB2973)
    - 800 °C, 1 h, standard inertion (oven B): 83 mg output (MB2979A)
    - 800 °C, 1 h, standard inertion: 104 mg output (MB2979B)

- 2.5 g of biochar lots MB2959/2956 milled with 3.8 g of KOH for 10/20 min and used for four batches (pellet pressing experiment)
    - 0.6 g pellet, 20 min milling time, 800 °C, 1 h, standard inertion (oven B): 78 mg output (MB2980A)
    - 0.6 g pellet, 10 min milling time, 800 °C, 1 h, standard inertion (oven A): 95 mg output (MB2980B)
    - 1.3 g pellet, 20 min milling time, 800 °C, 1 h, standard inertion (oven B): 163 mg output (MB2980C)
    - 1.3 g pellet, 10 min milling time, 800 °C, 1 h, standard inertion (oven A): 232 mg output (MB2980D)

- 2.5 g of biochar lot MB2978 milled with 3.8 g of KOH for 10/20 min and used for ten batches
    - 800 °C, 1 h, 10 min milling time, standard inertion (oven A): 95 mg (KJo-0165A)
    - 800 °C, 1 h, 20 min milling time, inertion potentially nonstandard (oven B): 146 mg (KJo-0165B)
    - 800 °C, 1 h, 10 min milling time, 10 °C/min, standard inertion (oven A): 125 mg (KJo-0166A)
    - 800 °C, 1 h, 20 min milling time, 10 °C/min, standard inertion (oven B): 114 mg (KJo-0166B)
    - 800 °C, 4 h, 10 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 102 mg (KJo-0167A)
    - 800 °C, 4 h, 20 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 61 mg (KJo-0167B)
    - 900 °C, 2 h, 10 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 23 mg (KJo-0168A)
    - 900 °C, 1 h, 20 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 89 mg (KJo-0168B)
    - 800 °C, 1 h, 10 min milling time, 3 °C/min, 2.0 g pellet pressed, standard inertion (oven B): 232 mg (KJo-0169)
    - 800 °C, 1 h, 20 min milling time, 3 °C/min, 2.7 g pellet pressed, standard inertion (oven B): 619 mg (KJo-0170)

- 5 g of biochar lot MB2978 milled with 7.5 g of KOH for 5 min and used for two batches
    - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven B): 642 mg (KJo-0171A)
    - 800 °C, 1 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 128 mg (KJo-0171B)

- 2.5 g of biochar lot MB2978 milled with 2.5 g of KOH for 2.5 min and used for two batches
    - 800 °C, 1 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 243 mg (KJo-0172A)
    - 800 °C, 2 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 169 mg (KJo-0172B)

- 3.0 g of biochar lot MB2978 milled with 4.5 g of KOH for 5 min and used for three batches
  - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven A), quartz tube shattered: 710 mg output (KJo-0175)
  - 800 °C, 1 h, 1.5 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 104 mg output (KJo-0176)
  - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven A): 908 mg output (KJo-0177)

- Various combined lots of milled biochar/KOH (1:1.5, lots MB2955/2974/2975/2976) used for two production experiments
  - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, questionable inertion (oven A): 549 mg output (KJo-0178)
  - 800 °C, 1 h, 3 °C/min, 6 x 0.6 g pellets pressed, standard inertion (oven B): 441 mg output (MB2981)

- Various combined lots of biochar (MB2946/2948/2949/2950/2951/2953/2956/MRa255) milled with KOH (1:1.5) – mixture heated itself moderately during milling
  - 800 °C, 1 h, 3 °C/min, 1.8 + 2.1 g pellets pressed, standard inertion (oven A): 587 mg output (TB1138)
  - 800 °C, 1 h, 3 °C/min, 1.8 + 1.7 g pellets pressed, standard inertion (oven B): 547 mg output (MRa320)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.8 g pellets pressed, standard inertion (oven B): 476 mg output (MB2984)
  - 800 °C, 1 h, 3 °C/min, 1.3 + 1.4 g pellets pressed, standard inertion (oven A): 395 mg output (TB1141)

- 8.5 g biochar (KJo-0173) milled with KOH (1:1.5) – brown powder, quickly turned dark brown, somewhat chunky and hot (steaming) upon contact with air, product thus not perfectly homogeneous
  - 800 °C, 1 h, 3 °C/min, 2 x 1.8 g pellets pressed, standard inertion (oven A): 538 mg output (TB1139)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 464 mg output (TB1140A)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven B): 480 mg output (TB1140B)

- 5.0 g biochar (KJo-0173) milled, sonicated (1 h), milled with KOH (1:1.5) – brown powder, no noteworthy observations
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A): 464 mg output (MRa323)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): output tbd (MRa326)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): output tbd (MRa327)

- Total quantity of material with sufficient quality (judging by SEM data) available: 7.4 g

- MB2976 A/B:	approx. 100 mg
- MB2980 A:	approx. 50 mg
- MB2980 C/D:	approx. 300 mg
- KJo-0166 B:	approx. 100 mg
- KJo-0167 B:	approx. 50 mg
- KJo-0168 B:	approx. 70 mg
- KJo-0171 A:	approx. 600 mg
- KJo-0171 B:	approx. 100 mg
- KJo-0172 A:	approx. 220 mg
- KJo-0175:	approx. 600 mg
- KJo-0176:	approx. 80 mg
- KJo-0177:	approx. 800 mg
- KJo-0178:	approx. 500 mg
- MB2981:	approx. 400 mg
- TB1138:	approx. 500 mg
- MRa320:	approx. 500 mg
- MB2984:	approx. 400 mg
- TB1139:	approx. 500 mg
- TB1140A:	approx. 400 mg
- TB1140B:	approx. 400 mg
- MRa323:	approx. 400 mg
- TB1141:	approx. 300 mg

4. Graphene analysis

- SEM (and partially Raman) results available for
  - TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A, MB2955A2, TB1136, MB2962A/B, MB2963A/B, TB1137-1/2/3 (step 2 products), MB2965B, MB2965B2, MB2966A/B, MB2967A/B, MB2970A/B, MB2971A/B, MB2972A/B, MB2973A/B, MB2974A/B, MB2975A/B, MB2976A/B, MB2979A/B, MB2980A/B/C/D, KJo-0165A/B, KJo-0166A/B, KJo-0167A/B, KJo-0168A/B, KJo-0169, KJo-0170, KJo-0171A/B, KJo-0172A/B, KJo-0175, KJo-0176, KJo-0177, KJo-0178, KJo-0171A GR, MB2981A, TB1138, MB2984, MRa320, TB1139, TB1140A, TB1140B, MB2988, MRa323 (step 2 products)
  - 900561-500MG (Sigma-Aldrich reference)
  - 924458-500MG (Sigma-Aldrich reference "Single-layer graphene sheets for battery, Bio-sourced")

- TEM results available for
  - MB2955A2 (step 2 product)

5. Currently open questions

Experiment overview – Step 1

Experiment	Reactor	Raw material		Acid				T	t	pressure			Wash		Output	
MB2928	AV1	6.0 g	BAFA neu Hemp Fibre VF	100.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	10.4 bar	742 g	Water	2.6 g	
Comment	Fibres not completely covered by liquid															
MB2933	AV1	6.1 g	BAFA neu Hemp Fibre VF	174.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	24 h	8.7 bar	->	9.5 bar	647 g	Water	1.2 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2935	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.9 g	0.19%	0.02 M	Sulfuric acid	180 °C	23 h	8.8 bar	->	9.5 bar	500 g	Water	1.3 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2936	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.8 bar	->	10.6 bar	589 g	Water	0.8 g	Combined and homogenized as lot no MRa231
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa225	AV1	6.0 g	BAFA neu Hemp Fibre VF	153.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	?	->	10.5 bar	500 g	Water	0.9 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa228	AV1	6.3 g	BAFA neu Hemp Fibre VF	154.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	11.0 bar	500 g	Water	0.9 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa231	AV1	6.1 g	BAFA neu Hemp Fibre VF	168.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.0 bar	->	10.5 bar	500 g	Water	0.8 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2937	AV1	9.4 g	Canadian Rockies Hemp	179.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	10.9 bar	569 g	Water	1.6 g	Combined and homogenzied as lot no MB2946X
Comment	very low amount of foreign material observed and removed during filtration															
MB2938	AV1	8.9 g	Canadian Rockies Hemp	175.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.2 bar	->	13.2 bar	577 g	Water	1.4 g	
Comment	very low amount of foreign material observed and removed during filtration															
MB2940	AV1	9.2 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	10.6 bar	555 g	Water	1.5 g	
Comment	very low amount of foreign material observed and removed during filtration															
MB2942	AV1	8.9 g	Canadian Rockies Hemp	177.6 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.3 bar	->	13.5 bar	578 g	Water	1.6 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2943	AV1	9.8 g	Canadian Rockies Hemp	178.2 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.3 bar	->	12.7 bar	611 g	Water	1.7 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2944	AV1	9.0 g	Canadian Rockies Hemp	178.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.4 bar	->	9.7 bar	593 g	Water	1.8 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2945	AV1	9.1 g	Canadian Rockies Hemp	177.7 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.7 bar	->	9.8 bar	6 g	Water	1.6 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2948	AV1	9.4 g	Canadian Rockies Hemp	175.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	25 h	9.0 bar	->	12.6 bar	475 g	Water	1.7 g	
Comment	low amount of foreign material observed and removed during filtration															

Experiment	Reactor	Raw material		Acid				T	t	pressure			Wash		Output
MB2949	AV1	9.2 g	Canadian Rockies Hemp	174.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.4 bar	->	10.2 bar	538 g	Water	1.8 g
Comment	low amount of foreign material observed and removed during filtration														
MB2950	AV1	9.1 g	Canadian Rockies Hemp	170.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h		->	11.1 bar	534 g	Water	1.6 g
Comment	low amount of foreign material observed and removed during filtration														
MB2951	AV1	9.3 g	Canadian Rockies Hemp	178.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	9.1 bar	->	11.2 bar	575 g	Water	1.8 g
Comment	low amount of foreign material observed and removed during filtration														
MRa255	AV1	9.2 g	Canadian Rockies Hemp	170.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.1 bar	->	12.8 bar	488 g	Water	1.7 g
Comment	some foreign material observed and removed during filtration														
MB2953	AV1	9.7 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.1 bar	->	10.4 bar	558 g	Water	1.8 g
Comment	low amount of foreign material observed and removed during filtration														
MB2956	AV1	9.8 g	Canadian Rockies Hemp	172.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	12.5 bar	548 g	Water	1.9 g
Comment	low amount of foreign material observed and removed during filtration														
MB2957	AV1	9.1 g	Canadian Rockies Hemp	176.8 g	1.50%	0.15 M	Sulfuric acid	180 °C	23.5 h	8.8 bar	->	8.5 bar	589 g	Water	1.5 g
Comment															
MB2958	AV1	9.3 g	Canadian Rockies Hemp	177.0 g	2.00%	0.20 M	Sulfuric acid	180 °C	24 h	8.9 bar	->	8.7 bar	562 g	Water	1.8 g
Comment															
MB2959	AV5	76.0 g	Canadian Rockies Hemp	953.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	25 h				3957 g	Water	14.9 g
Comment	low amount of foreign material observed and removed during filtration														
MB2960	AV1	3.8 g	Canadian Rockies Hemp (cut)	151.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.8 bar	->	10.5 bar	300 g	Water	0.6 g
Comment	Reaction performed with stirring														
MB2961	AV1	9.9 g	Canadian Rockies Hemp	172.0 g	5.00%	0.51 M	Sulfuric acid	180 °C	24 h	10.1 bar	->	16.1 bar	563 g	Water	1.7 g
Comment	low amount of foreign material observed and removed during filtration														
MB2964	AV1	9.0 g	Canadian Rockies Hemp	211.0 g + 140.8 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	8.8 bar	310 g + 603 g	Water	1.3 g
Comment	Hemp boiled with 1% H_2SO_4 at 95 °C, filtered and washed, then heated in Autoclave with fresh acid														
MB2978	AV5	76.3 g	Canadian Rockies Hemp	822.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	22 h				4112 g	Water	15.4 g
Comment															
KJo-0173	AV5	77.1 g	Canadian Rockies Hemp	942.1 g	1.00%	0.10 M	Sulfuric acid	185 °C	24 h				4283 g	Water	16.5 g
Comment															
KJo-0174	AV5	77.1 g	Canadian Rockies Hemp	945.0 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4384 g	Water	15.8 g
Comment															
MB2982	AV5	77.1 g	Canadian Rockies Hemp	999.5 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4258 g	Water	14.9 g
Comment															
MB2983	AV5	80.5 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4232 g	Water	16.6 g
Comment															

Experiment	Reactor	Raw material		Acid				T	t	pressure			Wash		Output
MB2985	AV5	79.5 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	200 °C	24 h				4241 g	Water	
Comment															
MB2986	AV5	79.9 g	Canadian Rockies Hemp	999.9 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h				4128 g	Water	16.0 g
Comment															

Experiment overview – Step 2

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
TB1131	A	1.2 g	MB2933	1.2 g	KOH	(90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.16 g
Comment		ground biochar: brown powder, strong electrostatic charge										(+ water)		(atm. Pressure)		
TB1132	A	1.3 g	MB2935	1.3 g	KOH	(90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder										(+ water)		(atm. Pressure)		
TB1133	A	1.0 g	MRa231	1.0 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1134	A	1.0 g	MRa231	1.0 g	KOH	(90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2946	A	1.0 g	MB2946X	1.0 g	KOH	(90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2947A	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g	10% HCl	40 °C	N2 stream	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		210 mbar		
MB2947B	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g	10% HCl	40 °C	N2 stream	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		210 mbar		
MB2947C	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h	Batch discarded				
Comment		ground biochar: brown powder; see temperature trend														
MB2947D	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g	10% HCl	100 °C	N2 stream	38 mg
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1135-1	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1135-2	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2952B	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2952C	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	3 h	10 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2952C2	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2955A	A	0.25 g	MB2946X	0.38 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	7 mg
Comment		Quartz tube shattered during experiment										(+ water)		(atm. Pressure)		
MB2955A2	A	0.25 g	MB2946X	0.38 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1136	A	0.26 g	MB2946X	0.39 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	14 g	10% HCl	100 °C	N_2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2962A	A	0.25 g	MB2946X	0.50 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2962B	A	0.25 g	MB2946X	0.50 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2963A	A	0.22 g	MB2960	0.22 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2963B	A	0.29 g	MB2960	0.29 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.12 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-1	A	0.26 g	MB2946X	1.06 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	23 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-2	A	0.28 g	MB2946X	1.13 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	23 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-3	A	0.26 g	MB2946X	1.06 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	23 g	10% HCl	100 °C	N_2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2965A	A	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N_2 stream	0.02 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		
MB2965B	A	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N_2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		
MB2965B2	B	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N_2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2966A	B	0.26 g	MB2959	1.04 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2966B	A	0.26 g	MB2959	1.04 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2967A	B	0.26 g	MB2964	1.04 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2967B	A	0.26 g	MB2964	1.04 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2970A	B	0.24 g	MB2957	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2970B	A	0.24 g	MB2957	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.02 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2971A	B	0.24 g	MB2958	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2971B	A	0.24 g	MB2958	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2972A	B	0.24 g	MB2961	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2972B	A	0.24 g	MB2961	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2973A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2973B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2974A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2974B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2975A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2975B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2976A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend										(+ water)		(atm. Pressure)		
MB2976B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.11 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend										(+ water)		(atm. Pressure)		
MB2979A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2979B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2980A	B	0.24 g	MB2959/2956	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980B	A	0.24 g	MB2959/2956	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980C	B	0.52 g	MB2959/2956	0.78 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	21 g	10% HCl	100 °C	N2 stream	0.16 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980D	A	0.52 g	MB2959/2956	0.78 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	21 g	10% HCl	100 °C	N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0165A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0165B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.15 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0166A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	10 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.13 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0166B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	10 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.11 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
KJo-0167A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	4 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0167B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	4 h	11 g	10% HCl	100 °C	N2 stream	0.06 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0168A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	900 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.02 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0168B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	900 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0169	B	0.80 g	MB2978	1.20 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	30 g	10% HCl	100 °C	N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0170	B	1.07 g	MB2978	1.61 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	38 g	10% HCl	100 °C	N2 stream	0.62 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0171A	B	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.64 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
KJo-0171B	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.13 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0172A	B	0.50 g	MB2978	0.50 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.24 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0172B	A	0.50 g	MB2978	0.50 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.17 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0175	A	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g	10% HCl	100 °C	N2 stream	0.71 g
		ground biochar (brown powder) compacted to two pellets of equal size, quartz tube shattered										(+ water)		(atm. Pressure)		
KJo-0176	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (5 min)	N2	1.5 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0177	A	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g	10% HCl	100 °C	N2 stream	0.91 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
KJo-0178	A	1.60 g	MB2955/74/75/76	2.40 g	KOH	(90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.55 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
MB2981	B	1.44 g	MB2955/74/75/76	2.16 g	KOH	(90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.44 g
		ground biochar (brown powder) compacted to six pellets of equal size										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
TB1138	A	1.54 g	various	2.31 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	59 g	10% HCl	100 °C	N2 stream	0.59 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
MRa320	B	1.43 g	various	2.14 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	53 g	10% HCl	100 °C	N2 stream	0.55 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
MB2984	B	1.40 g	various	2.10 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	53 g	10% HCl	100 °C	N2 stream	0.48 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
TB1139	A	1.39 g	KJo-0173	2.08 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	50 g	10% HCl	100 °C	N2 stream	0.54 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
TB1140A	A	1.28 g	KJo-0173	1.92 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	49 g	10% HCl	100 °C	N2 stream	0.46 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
TB1140B	B	1.28 g	KJo-0173	1.92 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	50 g	10% HCl	100 °C	N2 stream	0.48 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
MRa323	A	1.06 g	KJo-0173	1.60 g	KOH	(90%)	mill (3 min)	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.44 g
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
TB1141	A	1.08 g	various	1.62 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	30 g	10% HCl	100 °C	N2 stream	0.40 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
MRa326	A	1.28 g	KJo-0173	1.92 g	KOH	(90%)	mill (3 min)	N2	3 °C/min	800 °C	1 h	48 g	10% HCl	100 °C	N2 stream	
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
MRa327	B	1.04 g	KJo-0173	1.56 g	KOH	(90%)	mill (3 min)	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		